

The Empty Set

- Also known as the null set. This set has no elements.
- Denoted by $\emptyset$ and $\{\}$
- For instance, the set of all positive integers that are greater than their squares is the null set.
- A set with one element is called a singleton set.
- Empty set $\rightarrow \emptyset$ Singleton set $\rightarrow \{\emptyset\}$.

Naive Set Theory

- Note that the term object has been used in the definition of a set, Definition 1, without specifying what an object is.
- This description of a set was introduced by Cantor.
- The theory that results from this intuitive definition of a set, and the use of the intuitive notion that for any property whatever, there is a set consisting of exactly the objects with this property, leads to paradoxes, or logical inconsistencies. This was shown by B. Russell.
- These logical inconsistencies can be avoided by building set theory beginning with axioms. For Further Reading on Axiomatic Set Theory, Read the Book "Sets, Logic and Axiomatic Theories" by R.R. Stoll.
- In this book, we shall use Cantor's original set theory or Naive Set Theory.

2.1.2 Venn Diagrams

- Sets can be represented graphically using Venn diagrams.
- In Venn diagrams, the universal set U , which contains all the objects under consideration is represented by a rectangle.
- Circles or other geometrical figures are used to represent sets.
- Points are used to represent the particular elements of the set.

Example 7: Draw a Venn diagram that represents V , the set of vowels in the English alphabet.

Solution: We draw a rectangle to indicate the universal set U , which is the set of the 26 letters of the English alphabet. Inside this rectangle we draw a circle to represent V . Inside this